SIMATS ENGINEERING
ASSESSMENT TOOL 2(MEDIUM)

COURSE NAME: Ethical Hacking COURSE CODE: ITA14

COURSE OUTCOME COVERED

CO3: Analyze network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BT4)
Assessment Tool Weightage: 30%

Duration: 90 min Total Marks: 50

Port Scanning Challenge

Using Nmap, perform a security assessment on the server 192.168.1.20. Identify all open ports and running services. Analyze the purpose of each discovered service and recommend security measures to reduce the attack surface of the system.

Using Nikto, assess the web server hosted on 192.168.1.30. Identify potential web server misconfigurations, outdated components, or exposed files and directories. Discuss the security risks associated with your findings and suggest appropriate mitigation strategies.

Using Netcat (nc), investigate the host 192.168.1.40 by connecting to open ports and collecting available service banners. Analyze the information obtained and explain how attackers could use banner information during the reconnaissance phase of an attack.

Using Telnet, examine the server 192.168.1.50 to determine whether remote administration services are accessible. Identify any security concerns related to the discovered services and recommend more secure alternatives where appropriate.

Using Netdiscover, perform host discovery on the network segment 192.168.1.1–50. Identify active hosts on the network and analyze how the discovered information could assist an attacker in planning further enumeration or exploitation activities..

Code of Conduct: *I Dinesh R certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*
Signature of the student: Dinesh R

1. Security Assessment Using Nmap (192.168.1.20)

Aim

To identify open ports and running services using Nmap and recommend security measures.

Command

nmap -sV 192.168.1.20

To scan all TCP ports:

nmap -p- -sV 192.168.1.20

Analysis

Port	Service	Purpose	Risk
21	FTP	File transfer	Unencrypted data
22	SSH	Secure remote access	Weak passwords
80	HTTP	Web hosting	Unencrypted communication
443	HTTPS	Secure web access	Incorrect TLS configuration

These are common examples; actual open ports must be confirmed by scanning.

Security Recommendations

1. Disable unnecessary services.
2. Close unused ports using firewall rules.
3. Use SFTP instead of FTP.
4. Use strong passwords and SSH keys.
5. Update the operating system and services regularly.
6. Monitor logs for suspicious activity.

Result

Nmap helps identify exposed services and reduce the system's attack surface.

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2. Web Server Assessment Using Nikto (192.168.1.30)

Aim

To identify web server misconfigurations, outdated components, and potentially exposed files using Nikto.

Command

nikto -h http://192.168.1.30

For an HTTPS server:

nikto -h https://192.168.1.30

Analysis

Nikto checks for common web server security issues, including:

- Outdated server software.
- Missing security headers.
- Default files and configurations.
- Potentially exposed directories.
- Dangerous or unnecessary HTTP methods.

Security Risks

Finding	Risk	Mitigation
Outdated software	Known vulnerabilities	Update software
Missing security headers	Increased exposure to certain attacks	Configure suitable headers
Directory listing enabled	Information disclosure	Disable directory listing
Exposed configuration files	Leakage of sensitive information	Restrict access and remove unnecessary files
Unnecessary HTTP methods	Potential misuse	Disable unused methods

Security Recommendations

1. Update the web server and its components.
2. Remove default files and unnecessary directories.
3. Disable directory listing.
4. Configure appropriate HTTP security headers.
5. Restrict access to sensitive files.
6. Use HTTPS with valid TLS configuration.

Result

Nikto helps identify common web server weaknesses and supports security hardening.

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3. Service Banner Analysis Using Netcat (192.168.1.40)

Aim

To connect to accessible services and collect banner information using Netcat.

Command

nc -nv 192.168.1.40 22

For an HTTP service:

nc -nv 192.168.1.40 80

To send a basic HTTP request:

GET / HTTP/1.1

Host: 192.168.1.40

Analysis

Service banners may reveal:

- Service names.
- Software versions.
- Server type.
- Operating system clues.
- Protocol information.

Security Risks

Attackers may use banner information to identify outdated software, research known vulnerabilities, and select possible attack methods.

Security Recommendations

1. Update services regularly.

2. Avoid exposing unnecessary services.
3. Hide excessive version information where practical.
4. Restrict access using firewall rules.
5. Monitor suspicious connection attempts.

Result

Netcat helps identify service banners and provides useful information for security assessment.

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4. Remote Administration Assessment Using Telnet (192.168.1.50)

Aim

To check whether remote administration services are accessible and identify related security risks.

Command

telnet 192.168.1.50 23

To check whether SSH is accessible:

telnet 192.168.1.50 22

Note: Connecting to port 22 using Telnet only checks the TCP connection; it does not perform an SSH login.

Analysis

Service	Port	Purpose	Security Concern
Telnet	23	Remote administration	Transmits data without encryption
SSH	22	Secure remote administration	Weak credentials or poor configuration
HTTP	80	Web management	Unencrypted communication
HTTPS	443	Secure web management	Incorrect TLS configuration

These are common ports to check, not confirmed findings.

Security Risks

- Telnet may expose usernames and passwords.
- Weak authentication may allow unauthorized access.
- Exposed management services increase the attack surface.
- Poor access controls may allow unauthorized administration.

Security Recommendations

1. Disable Telnet.
2. Use SSH for secure remote administration.
3. Apply strong passwords and multifactor authentication where supported.
4. Restrict management access to trusted IP addresses.
5. Monitor login attempts.
6. Update server software regularly.

Result

Telnet connectivity checks help identify accessible TCP services, while secure alternatives reduce the risk of unauthorized access.

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5. Host Discovery Using Netdiscover (192.168.1.1–50)

Aim

To identify active hosts on the local network using Netdiscover.

Command

sudo netdiscover -r 192.168.1.0/24

To use a smaller range where supported:

sudo netdiscover -r 192.168.1.1-50

Note: Netdiscover commonly uses CIDR network ranges. If the second command is unsupported, use the appropriate network range or interface settings for your version.

Analysis

Netdiscover may identify:

- Active IP addresses.
- MAC addresses.
- MAC address vendor information.
- Devices communicating on the local network.

Security Risks

An attacker could use discovered hosts to map the network, identify potential targets, and perform further enumeration. Exposed or poorly secured devices may increase the risk of unauthorized access.

Security Recommendations

1. Use network segmentation.
2. Restrict unauthorized device connections.
3. Secure routers and switches.
4. Monitor unknown devices.
5. Use strong authentication for network equipment.
6. Disable unnecessary services on connected devices.

Result

Netdiscover helps identify active devices and supports network inventory and security assessment.